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Agriculture

Abstract

There are significant issues associated with agriculture and its relationships with greenhouse gas emissions and air quality. Just as in other areas, energy is needed for farming, and transitioning to renewable energy in agriculture is an important goal. Because farms are large in area, the electrical grid has higher costs of transmission and distribution in rural parts of the world. There are many places where there is no electrical grid serving farm populations. The decreases in the costs of wind- and solar-generated electricity and the progress in battery development are very beneficial for agriculture and off-grid electrical systems. Sustainable development in agriculture is a high priority and has great value for society. Nutritious food is one of the most important needs for good health and well-being. Food security for all people is one of the Sustainable Development Goals. The United Nations Food and Agriculture Organization is providing leadership for sustainable agriculture.

12.1 Introduction

There are many topics related to reducing greenhouse gas emissions that are associated with agriculture in particular. Agriculture connects closely with sustainability issues having to do with land use, water use, and food security/world hunger issues. At the same time, there are other issues, already discussed more generally, that are relevant for agriculture as well: Renewable energy generation, electrification of transportation (here including farm machinery), and a smarter infrastructure.

The generation of electricity using wind and solar energy for farming with electric tractors has a bright future. The cost of solar panels continues to decrease, and battery storage of renewable electricity has become competitive with other alternatives (see Chapter 8). The progress in battery developments is very important for agriculture because battery purchase prices below \$50/kWh, combined with decreasing costs for electricity from wind and solar energy, will bring low energy prices to agriculture.

The electrification of transportation within agriculture includes electric tractors and trucks, as well as cars. There is progress with respect to the development of new products and their use by farmers. The electrical energy for use on the farm can be produced by adding solar panels to farm buildings and generating electricity for local use, and battery storage can be used to manage supply and demand of electricity. Some farms are able to be connected to a local electrical grid, whereas others may need to be served by an off-grid system developed and operated by the farmer (see Chapter 9).

There are land use issues that arise in agriculture because of the competition between food production, crops intended for ethanol production, land devoted to wind and solar energy generation, and land used for the production of wood (paper) and cotton (fabrics). Some of these land use issues are less difficult than others. For instance, solar panels can often be placed on buildings, over parking lots, and on land that is not suitable for farming.

Much of the technical development for advancing sustainable agriculture has been achieved, and part of the challenge now is getting those developments recognized as having great value. Some of the acceptance process in agriculture relies on what farmers have learned by experimenting in their fields. Therefore, it is important that extension service professionals continue to distribute this valuable information. The Internet has also, of course, become a valuable library and information system that is being used in support of agriculture.

12.2 Sustainable Development Goals

As discussed in Chapter 2, the UN General Assembly (United Nations, 2015) adopted the resolution *Transforming Our World: The 2030 Agenda for Sustainable Development* on September 25, 2015. This resolution includes a set of 17 Sustainable Development Goals (SDGs) that are very high priorities for both the United Nations and the signatories of the Paris Climate Agreement. Many of these SDGs relate clearly to agriculture.

The second of these SDGs is to end hunger, achieve food security and improved nutrition, and promote sustainable agriculture. The sixth SDG includes sustainable management of water. The seventh SDG is to ensure access to sustainable energy for all. The fourteenth SDG includes sustainable use of marine resources. The fifteenth SDG promotes sustainable management and use of land and terrestrial ecosystems. The Food and Agriculture Organization (FAO) has identified principles and 20 actions in support of the 2030 Agenda for Sustainable Development (FAO, 2018). These 20 actions are strongly interrelated and are described as follows:

Action 1: Facilitate access to productive resources, finance and services.

Rural infrastructure, resources and services are important to support agriculture, and renewable electricity needs to be a part of rural infrastructure. The decrease in costs of solar panels and batteries is beneficial to agriculture, including those who must develop their own off-grid generating system.

Action 2: Connect smallholders to markets. There is significant value in having an infrastructure that allows those in agriculture to connect with people in communities that provide agricultural supplies and consumers of agricultural products (markets). Inexpensive electric vehicles (EVs) powered by solar energy can help rural residents connect to local communities in an inexpensive and sustainable way.

Action 3: Encourage diversification of production and income. Cultivating multiple crops has value for many producers and increases resilience in the face of weather, disease, or other adverse events. Additionally, having diverse income streams can mean employment beyond the farm for one or more members of a family, providing opportunities that often benefit the family. Again, an inexpensive EV has value for those seeking diversification.

Action 4: Build producers' knowledge and develop their capabilities. Education has great value for farmers, and there are many opportunities to learn from extension organizations, farm organizations, and others. The Internet has many sites that are beneficial for producers. It is beneficial for producers to have electricity, to be connected to the Internet, and to be able to have access to market information and weather forecasts.

Action 5: Enhance soil health and restore land. Any good farmer will tell you that good soil stewardship is necessary for success. Traditionally this has largely meant optimal use of fertilizer, but that is now just one aspect of soil management. Because of climate change, soil erosion associated with intense rainfall and flooding has become a greater problem than in the past. The FAO has voluntary guidelines for sustainable soil management (FAO, 2017) that include actions to minimize soil erosion, enhance soil organic matter content, foster soil nutrient balance, minimize soil salinization, acidification, and alkalinization, prevent soil contamination, enhance soil biodiversity, reduce soil compaction, and improve soil water management (FAO, 2017).

Action 6: Protect water and manage scarcity. Water quantity and water quality are central topics in agriculture because about 70% of all water is used for crops and livestock. Insufficient or polluted water can thus have tremendous impacts on food products, which then also have impacts on consumers' health.

Water scarcity is an issue in many locations where crop irrigation is an active process, using wells, canal systems, or both. As water becomes less prevalent, efforts to improve water efficiency will have more value and should be extended with further developments. Sustainable management of water resources more broadly is also important for society; it is needed to end hunger and it is the sixth SDG.

Action 7: Mainstream biodiversity conservation and protection of ecosystem functions. Healthy ecosystems are a valuable part of efforts to achieve food security and sustainable agriculture. This can include protecting not only wildlife in general, but also ecosystem biodiversity in places such as soil microorganisms. Healthy and diverse organism populations in soil are generally beneficial, and biodiversity in ecosystems has value in efforts to increase food production. International progress in programs to maintain important genetic resources for plant agriculture has been reported (FAO, 2015), and many programs exist to maintain genetic resources and to enable good ecosystem function in agriculture in many locations.

Action 8: Reduce losses, encourage reuse and recycling, and promote sustainable consumption. The issue of managing food products so that they are used effectively and efficiently is a substantial challenge because many foods may spoil. Somewhere around one-third of food production is lost due to poor food management. Packaging, storage conditions, managing supply and demand, preservation methods, and consumer education are aspects of programs to reduce food waste (FAO, 2011; Esguerra, 2018). Good management of food products has significant economic value and needs to be an important priority.

Action 9: Empower people and fight inequalities. One way to empower people is to educate them. Education is very important in agriculture because knowledge regarding farming practices, product management, renewable energy choices, water management, and soil stewardship are foundational to implementing these practices, and therefore these educational efforts have great value. Children that live in rural areas, of course, should go to school and receive an education. Part of that education should include information tailored to the situations those children are likely to encounter as they grow up and begin their adult careers.

Action 10: Promote secure tenure rights. The right to have access to land to farm it is an issue in some locations. Land ownership is a key factor because soil stewardship is often better when the person managing the land is the owner; they have more of an incentive to improve soil quality. One of the issues in ocean stewardship is the problem of managing the ocean as a common resource (see Chapter 15).

Action 11: Use social protection tools to enhance productivity and income.

Agriculture is a physically, cognitively, and socially demanding occupation. There are many risks for those who are working in production agriculture. Social protection programs reduce risks and have long-term value in agriculture. For instance, food availability for all has been demonstrated to work effectively and have positive results.

Action 12: Improve nutrition and promote balanced diets. Nutrition education is beneficial because it allows people to understand the health values associated with food choices. Having the opportunity to select nutritious foods for a diet is one part of the food pyramid, along with education about what food choices are better than others. The affordability of milk, meat, fruits, and vegetables impacts health for some because they do not have sufficient resources to purchase food for a good diet.

Action 13: Prevent and protect against shocks to enhance resilience.

There are a number of national disaster risks associated with agriculture: Floods, wildfires, and very dry conditions (including droughts). Resilience is important in agriculture because disasters such as these recurrently happen and those impacted often need help. Furthermore, because of climate change, there have been more of these types of disaster events recently. Good communication systems are valuable when timely information is needed.

Action 14: Prepare for and respond to shocks. There is significant value in having prepared for potential disasters, having emergency supplies, and having a plan to follow to address an emergency. Even basic and effective communication with respect to impending weather and wildfire events has great value and can save lives or livelihoods.

Action 15: Address and adapt to climate change. As mentioned before, agriculture is being impacted by climate change – in some ways more severely than other sectors of the population – in the form of more floods, wildfires, and droughts than in earlier years. The severity of these events is also exacerbated. This feeds back into the need to reduce greenhouse gas emissions associated with agriculture (for example, by using electricity from renewable sources, electrifying operations, and producing fertilizer with sustainable energy and electrolysis of water).

Action 16: Strengthen ecosystem resilience. Whether ecosystems are functioning, faltering, or failing, they are impacted by weather and climate change. Actions are often needed to restore ecosystems and improve their health and resilience after disasters strike. This includes updated evaluation methods that allow better assessments of lands and ecosystems to determine what actions are needed to improve the state of the land at the site of interest.

Action 17: Enhance policy dialogue and coordination. Government policy has been important in agriculture in many countries for many years (see Chapter 10). Research, education, and extension programs have provided support for agriculture since 1863 in the United States. There have been global cooperative efforts through the FAO for many years that have been very beneficial. Policy actions are almost always an important part of efforts to advance sustainable agriculture and reduce greenhouse gas emissions. Of particular importance currently is a more extensive dialogue on water policies in support of agriculture using renewable energy.

Action 18: Strengthen innovation systems. Innovation has always been very important in the success of agriculture in the past, and it is needed moving forward into the future to achieve the SDGs. The transformation to renewable energy, the electrification of agricultural operations, and the development of battery storage systems to support rural electrification all depend on further innovation. Innovation is also important in sustainable ecosystem management.

Action 19: Adapt and improve investment and finance. Investments in agriculture are necessary for reaching the SDG to end hunger and achieve food security for everyone. Land ownership is a large investment, as are the investments in farm equipment, seeds, and fertilizers that are needed for modern agriculture. These all require working capital. Additional investment is needed for solar panels and battery storage to have renewable electricity. A modern home with appliances and a communication network completes the picture of what is desired. A nearby community with schools, a hospital, stores, a bank, a post office, and a local government should be part of the rural infrastructure.

Action 20: Strengthen the enabling environment and reform the institutional framework. Rural development in support of sustainable agriculture is important. Support and policies implemented by local and state governments are significant, but other organizations are involved as well in many countries. For instance, there are global companies that have products such as tractors and farm equipment for sale, and these companies can play key roles in the electrification of that equipment. Education and the availability of information, naturally, should also be part of the enabling environment.

12.3 Greenhouse Gases and Agriculture

Agriculture has several important impacts on greenhouse gas emissions, some of which are more obvious than others. Equipment, transportation of products,

and nitrogen-based fertilizers contribute to emissions. But organic carbon in soils is also beneficial in many locations because crop yields are better in soils with higher concentrations of organic matter and good biodiversity. Thus, actions by farmers to increase soil organic matter and water infiltration into soils can often have positive results in environments where increasing soil moisture availability has benefits. Another less obvious implication for greenhouse gas emissions from agriculture is the potential for ocean-based farming.

The oceans have the potential to support the growth of algae, phytoplankton, and other photosynthetic microorganisms that fix carbon. One of the opportunities that exists is to enhance biomass production in the oceans and other locations. Food production in water environments could be increased substantially. There certainly are science and technology development challenges to farming the oceans, but there are also great opportunities to increase production of food from the oceans. Additionally, there are important social and policy issues to develop and put in place in order to allow ocean mariculture farmers to lease a plot in the ocean and responsibly develop it into a more productive ecosystem (EU, 2017). Progress is needed in managing the commons of the ocean and investing in developing seafood farming technologies for ocean environments. Increasing photosynthetic production in the ocean, however, can increase the harvest of useful products that are higher up on the food chain.

One option, for illustrative purposes, would be to develop a community of people living in houseboats on the ocean at a selected location with an approved global position and an approved geographical area to use productively. Minerals and fertilizers that are needed for primary production of algae and phytoplankton could be added to the water regularly to support photosynthetic growth. The organisms needed to have a productive ecosystem could be added at the site to develop a productive community. Seaweed and other plant communities could be established. Solar panels with battery storage could be used for electricity. The community could have a boat to take harvested products to market and bring supplies from land regularly. Delivery using drones might be possible as well. There is a large part of the ocean that needs to be fertilized to have productive primary growth of biomass. Nitrogen, phosphorus, iron, and silicon are the elements that have been identified as limiting nutrients for primary productivity in the ocean (Sigman and Hain, 2012). While about 80% of the surface of the Earth is covered by water, only about 50% of primary carbon fixation by photosynthesis occurs in the oceans.

This is not a far-off hypothetical scenario. There are some individuals that are farming the ocean and producing seaweed, kelp, and other sea vegetables using a three-dimensional underwater scaffolding support system (Schiffman, 2016). Oysters, mussels, scallops, and shellfish are harvested in the area of the growing plants. One of the challenges of ocean farming, though, is to manage the ocean farming volume in an environmentally positive way such that environmental quality improves – rather than degrades – because of the actions taken to make the location productive.

12.4 Renewable Fuels

Some renewable fuels are produced from crops and crop waste materials. Perhaps most notably, ethanol is produced in significant quantities in several countries. The most common substrates for ethanol fermentation in the United States are corn and grain sorghum. Sugarcane is a common substrate in Brazil, which produces ethanol in significant quantities. Cellulose has been investigated as a substrate, but the economics are not very good at the present time (2019). Waste carbohydrate sources, such as grain dust cereal wastes, can also potentially be used to produce ethanol.

Another renewable fuel option from agriculture is through anaerobic digestion of agricultural wastes to produce methane. Waste products with this potential include dairy manure, food manufacturing waste, feed waste, and food waste. The economics of methane production depend on the alternatives that are available, but where methane from natural gas is expensive, the process of anaerobic digestion can provide gas that has value as a renewable fuel.

12.5 Air Quality

Finally, the issue of good air quality can centrally include agricultural concerns. The burning of crop residues and pastures impacts air quality significantly where it is allowed. The use of solid fuels and fires for cooking and heating, common in underdeveloped rural areas that often overlap with agricultural areas, causes air pollution and impacts health.

Larger domesticated animal operations have a different set of air quality issues that can arise. Dust from large cattle (and other animal) feeding operations is an issue when the feedlot soil is dry. There are odors from animal feeding operations in many cases. Where animal feeding takes place inside buildings, indoor air quality may be a problem. There can also be an issue with microbial pathogens that may move through the air to contaminate other organisms.

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